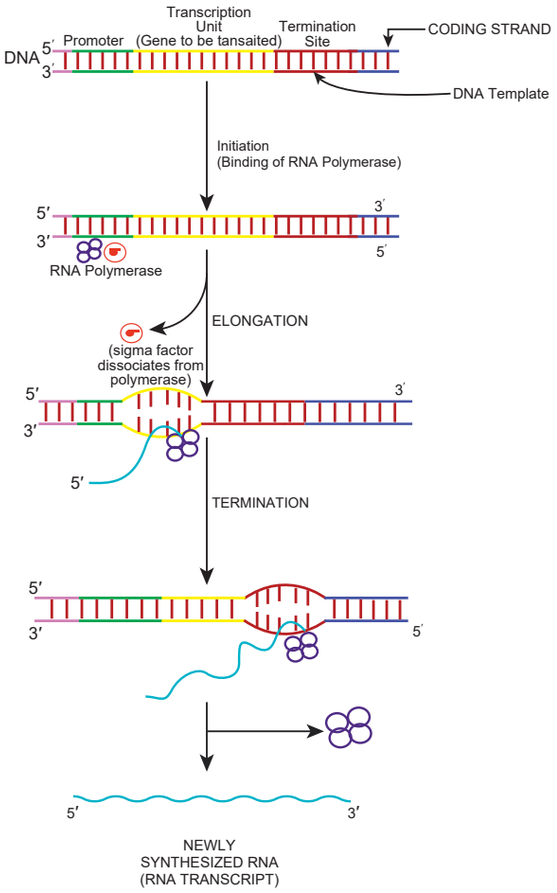


Figure 12.2: Major events in transcription

3. The first nucleoside triphosphate is placed at polymerization start site (near to the initial binding site) and synthesis begins.
4. RNA polymerase then moves along the DNA, adding ribonucleotides, to the growing RNA chain.
5. RNA polymerase reaches chain termination sequence and both the newly synthesized RNA and the polymerase are released. Two kinds of termination events are known those that are self – terminating (dependent on the base sequence only) and those that require the presence of the termination protein Rho.

Initiation of a second round of transcription need not await completion of the first, for the promoter becomes available once RNA polymerase has polymerized 50–60 nucleotides. In bacteria most mRNA molecules are degraded within a few minutes after synthesis. This degradation enables cells to dispense with molecules that are no longer needed.

In prokaryotes mRNA molecules commonly contain information for the amino acid sequences of several different polypeptide chains. In this case, such a molecule is called polycistronic mRNA. Cistron is a term used to mean a base sequence encoding a single polypeptide chain. The genes contained in polycistronic mRNA molecule (Figure 12.3) often encode the different portions of a metabolic pathway. For example, in *E. coli* the ten enzymes needed to synthesize histidine are encoded in one mRNA molecule.

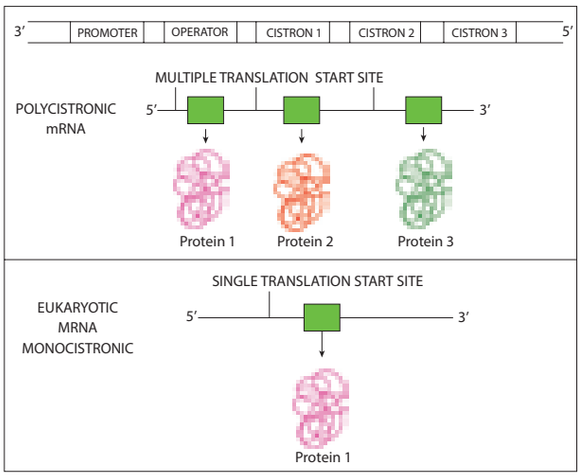


Figure 12.3: Polycistronic mRNA

In prokaryotes the immediate product of transcription (called the primary transcript) is mRNA, in contrast in eukaryotes the primary transcript must be converted to mRNA. This conversion called RNA processing consists of two types of events- modification of termini and excision of untranslated sequences (non-coding sequence or introns) embedded

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